

Aggregate Stagnation and Sectoral Divergence: Total Factor Productivity in Colombia, 2005–2024

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Abstract

Colombia’s near-zero aggregate total factor productivity (TFP) growth conceals economic activities moving in different directions. We reconstruct the long-run evolution of TFP for the total economy and nine activities using the annual production-account results published by the national statistical office (DANE). We chain the 20 annual log growth rates from 2005 to 2024, recover the annual aggregation weights implicit in the publication, validate them against nominal gross-output shares, and decompose aggregate TFP into exact activity contributions. The aggregate TFP index fell from 100 in 2004 to 98.5 in 2024, equivalent to an annualized rate of -0.073 percent. Trade TFP increased by 23.2 percent, while mining TFP fell by 32.1 percent. Four activities contributed 0.315 percentage points per year to aggregate TFP, but five activities subtracted 0.389 points. Thus, aggregate stagnation was the net result of offsetting positive and negative sectoral contributions. An accounting counterfactual that sets annual TFP growth to zero in the five activities with negative long-run performance raises production-account output growth from 3.26 to 3.65 percent per year. The exercise measures an accounting margin, not a causal policy effect or a counterfactual path for GDP.

Keywords: total factor productivity; growth accounting; KLEMS; sectoral productivity; Colombia.

JEL codes: O47; O54; E23; C43.

1 Introduction

Colombia’s near-zero aggregate total factor productivity (TFP) growth is not evidence that every economic activity stagnated. The same aggregate result can reflect either similarly weak outcomes across activities or large positive and negative sectoral contributions that

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offset one another. The distinction matters because the diagnosis changes: a common slow-down calls for examining economy-wide constraints, whereas offsetting contributions require identifying what happened within the activities on each side of the balance.

This paper asks whether Colombia’s aggregate TFP stagnation over the 20 annual growth rates from 2005 through 2024 was a common sectoral outcome or the net result of divergent activity paths. We use the production-account results published by the Departamento Administrativo Nacional de Estadística (DANE) for the total economy and nine broad economic activities (DANE, 2026a). The DANE framework follows the KLEMS growth-accounting approach: output growth is decomposed into the contributions of labor, capital, energy, materials, services, and a TFP residual (DANE, 2022; OECD, 2001; LA-KLEMS, 2021). The production approach provides a comparable activity history because intermediate inputs enter explicitly.

The paper makes three contributions. First, it converts DANE’s annual TFP growth rates into chained indices for each activity and the total economy. This permits a direct comparison between the levels of 2004 and 2024. Second, it recovers the annual aggregation weights implicit in the published tables, validates them against nominal gross-output shares, and uses them to obtain exact activity contributions to aggregate TFP. The recovery reproduces all 13 published aggregate columns to numerical precision. Third, it uses the same accounting identity to construct a transparent counterfactual. Annual TFP growth is set to zero in the five activities with negative long-run TFP growth, while all other input contributions and annual weights remain fixed. The same six CJC calculation sheets, included in both accompanying workbooks, show these calculations and map every figure to its source range.

The main result is that aggregate stagnation was real, but it was not a common sectoral pattern. The chained index for the total economy fell by 1.5 percent. Trade TFP increased by 23.2 percent, whereas mining TFP fell by 32.1 percent. Trade made the largest positive contribution to aggregate TFP, at 0.130 percentage points per year. Finance and real estate made the largest negative contribution, at -0.154 points, followed by mining at -0.115 points. Manufacturing and construction subtracted 0.058 and 0.054 points per year, respectively.

The accounting counterfactual illustrates the aggregate importance of the negative contributions. Production-account output grew by 3.26 percent per year. If annual TFP growth in mining, finance and real estate, construction, manufacturing, and electricity, gas, and water had been zero, the same accounting identity yields growth of 3.65 percent per year. The difference is 0.39 percentage points. This is not an estimate of a feasible policy intervention. It does not model changes in factor demand, prices, input use, or resource allocation, and it does not refer to GDP growth.

The rest of the paper is organized as follows. Section 2 presents the aggregate and sectoral results. Section 3 discusses interpretation, limitations, and policy questions. Section 4 concludes. The appendix reports supporting growth-accounting results, subperiod contributions, and the activity classification.

2 Results

2.1 Aggregate TFP ended below its 2004 level

Figure 1 reports the chained TFP index for the total economy. The index increased from 100 in 2004 to 100.1 in 2005 and 102.3 in 2006, then declined and remained below 100 from 2009 onward. It reached 98.5 in 2024. The cumulative decline was 1.5 percent, equivalent to an annualized log growth rate of -0.073 percent.

Chained TFP index for the total economy

2004 = 100; annual log growth rates from the DANE production approach

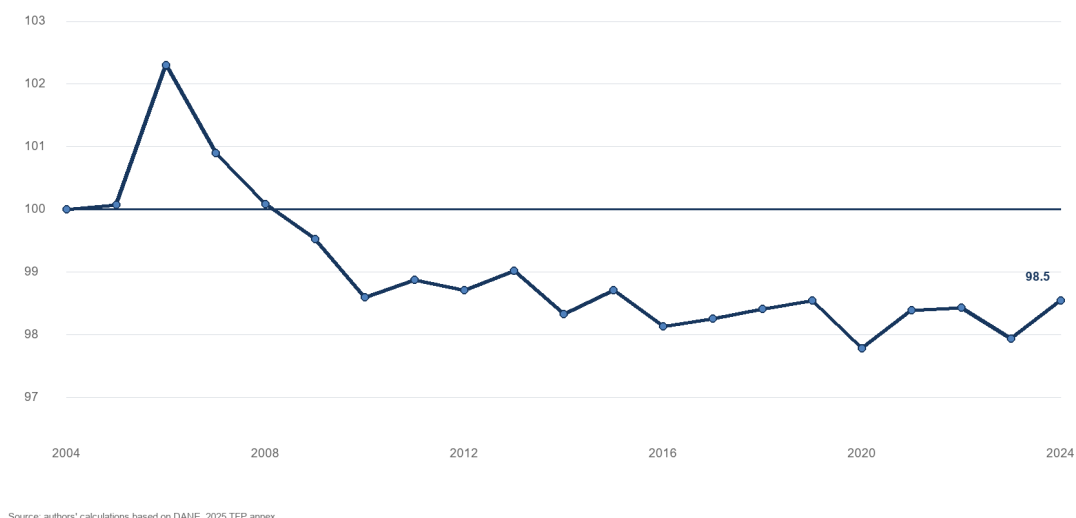


Figure 1: Chained TFP index for the total economy, 2004–2024

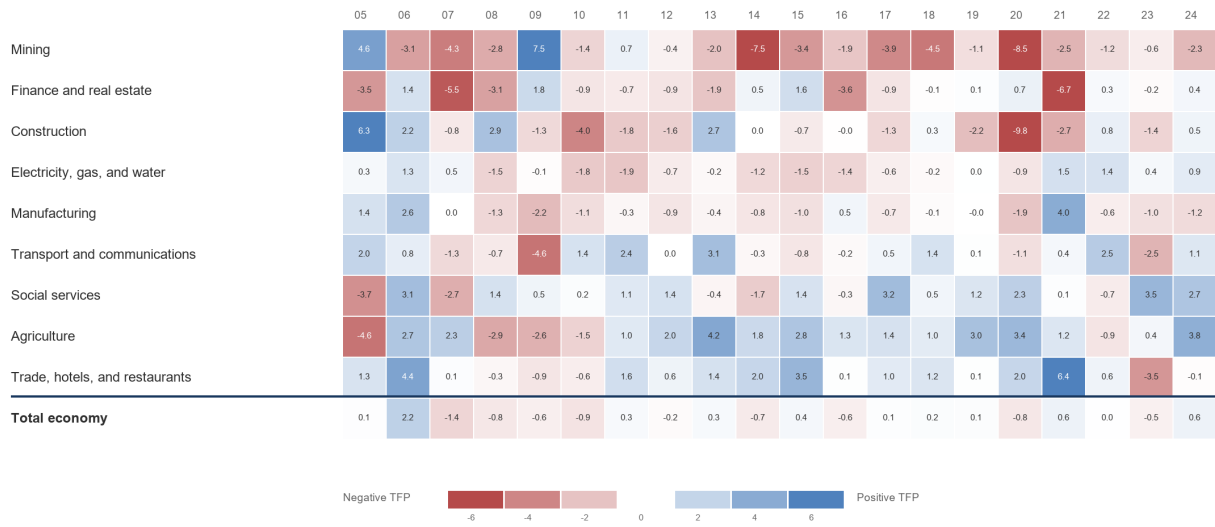
Most of the aggregate decline occurred before 2011. Aggregate TFP averaged -0.235 percent per year in 2005–2010. The subsequent averages were close to zero: 0.022 percent in 2011–2015, -0.040 in 2016–2019, and -0.001 in 2020–2024. The early deterioration stopped, but aggregate TFP did not enter a sustained growth path.

2.2 Aggregate stagnation concealed sectoral paths moving in opposite directions

Figure 2 displays the 20 annual TFP growth rates that determine the long-run results. The activities are ranked from the lowest to the highest cumulative TFP growth, with the total economy shown in the final row. Mining recorded negative TFP growth in 17 of 20 years and manufacturing in 15. Agriculture and trade recorded positive TFP growth in 15 years; social services did so in 14.

Annual TFP growth by activity

2005-2024, annual log growth rates (%); activities ranked by cumulative TFP growth



Note: the color scale is capped at +/-6 points; cell labels retain observed values.

Source: authors' calculations based on DANE, 2025 TFP annex.

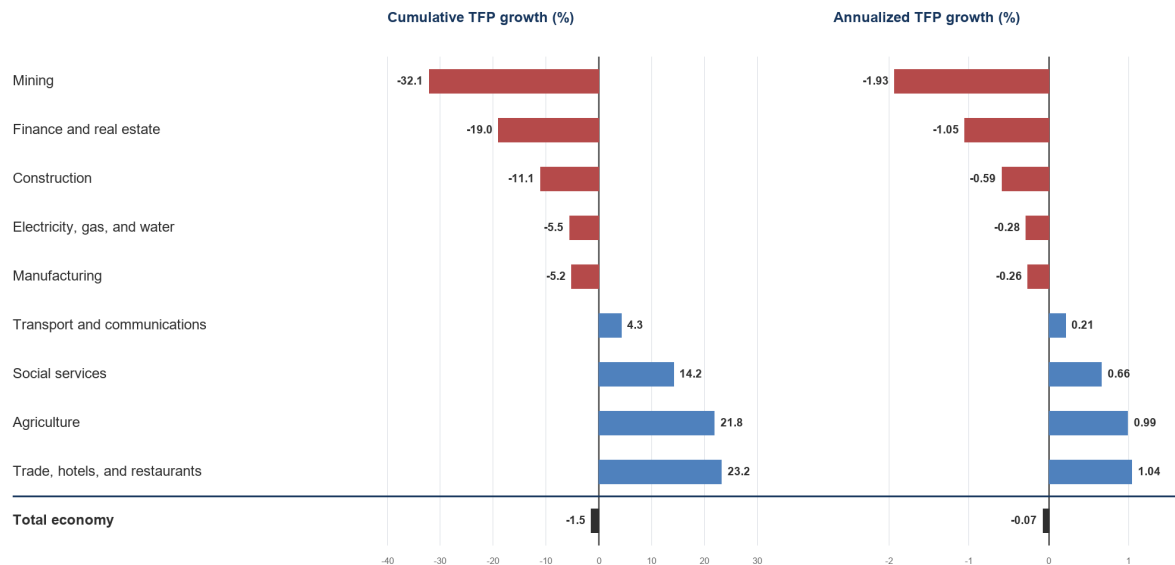
Figure 2: Annual TFP growth for the total economy and nine activities, 2005–2024

The annual observations also show why long-run averages should not be interpreted as smooth trends. Construction TFP fell by 9.8 percentage points in 2020. Trade TFP increased by 6.4 points in 2021, while finance and real estate fell by 6.7 points. These changes affect the full-period averages, but the sign frequencies show that the strongest long-run results are not produced by a single year alone.

Figure 3 reports the cumulative and annualized measures. Trade had the highest cumulative TFP growth, 23.2 percent, equivalent to an annualized rate of 1.04 percent. Agriculture increased by 21.8 percent and social services by 14.2 percent. Mining had the largest decline, 32.1 percent, equivalent to an annualized rate of -1.93 percent. Finance and real estate lost 19.0 percent and construction 11.1 percent.

Long-run sectoral TFP performance

20 annual rates, 2005-2024; the total economy is shown in dark gray



Source: authors' calculations based on DANE, 2025 TFP annex.

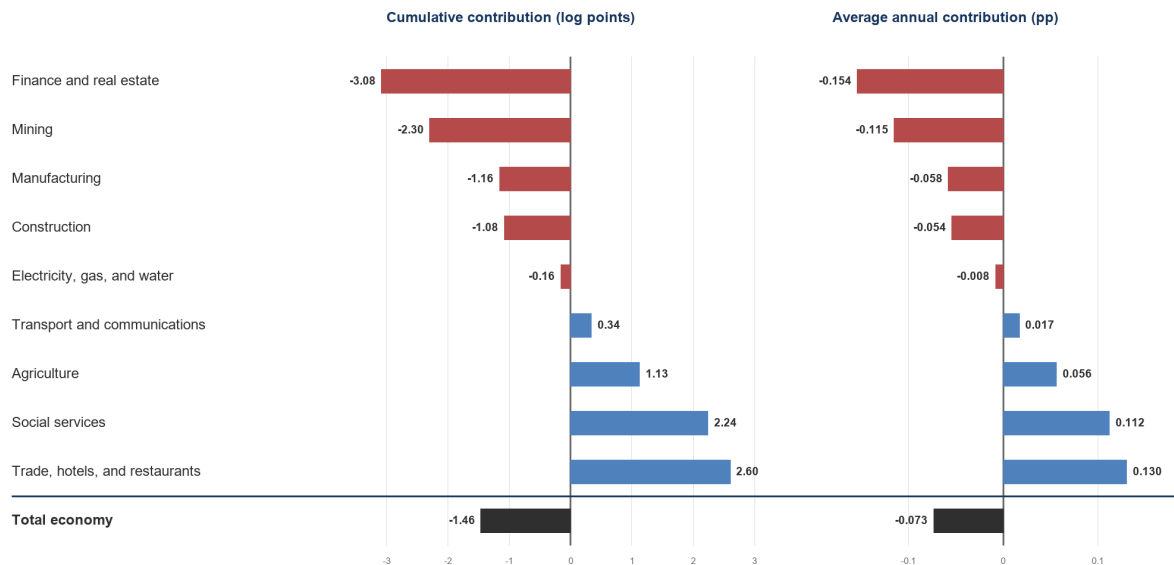
Figure 3: Cumulative TFP growth from 2004 to 2024 and annualized growth over 2005–2024

2.3 Positive and negative sectoral contributions nearly offset

Sectoral TFP growth does not determine an activity's aggregate contribution on its own. The contribution also depends on its annual aggregation weight. Figure 4 and Table 1 report both dimensions. Four activities contributed 6.308 cumulative log points, or 0.315 percentage points per year. Five activities subtracted 7.770 cumulative log points, or 0.389 points per year. The net contribution was -1.462 log points, equivalent to the aggregate annualized TFP growth rate of -0.073 percent.

Activity contributions to aggregate TFP

2005-2024; the total economy is shown in dark gray



Source: authors' calculations based on DANE, 2025 TFP annex.

Figure 4: Activity contributions to aggregate TFP, 2005–2024

Table 1: Sectoral TFP performance and contribution to the total economy, 2005–2024

Activity	Average weight (%)	Cumulative TFP growth, 2004–24 (%)	Annualized TFP growth (%)	Contribution (pp per year)
Trade, hotels, and restaurants	12.98	23.17	1.04	0.130
Social services	16.05	14.16	0.66	0.112
Agriculture	6.11	21.84	0.99	0.056
Transport and communications	8.63	4.26	0.21	0.017
Electricity, gas, and water	3.42	-5.53	-0.28	-0.008
Construction	8.64	-11.06	-0.59	-0.054
Manufacturing	23.43	-5.15	-0.26	-0.058
Mining	6.08	-32.09	-1.93	-0.115
Finance and real estate	14.66	-19.01	-1.05	-0.154
Total economy	100.00	-1.45	-0.07	-0.073

Notes: Cumulative growth compares the 2024 chained index with 2004. Annualized TFP is the mean of the 20 annual log growth rates from 2005 through 2024. Sectoral cumulative and annualized TFP describe performance within each activity and do not add across activities. The contribution multiplies annual activity TFP by its annual weight before averaging; the nine contributions add to aggregate TFP. Figures may not add because of rounding.

Source: Authors' calculations based on DANE, 2025 TFP annex.

Trade made the largest positive contribution, at 0.130 percentage points per year. Its

average weight was 13.0 percent and its own TFP increased at an annualized rate of 1.04 percent. Social services contributed 0.112 points. Agriculture had a lower average weight of 6.1 percent, so its contribution was 0.056 points. Transport contributed 0.017 points.

Finance and real estate made the largest negative contribution, at -0.154 percentage points per year. Its own TFP fell at an annualized rate of 1.05 percent and its average weight was 14.7 percent. Mining subtracted 0.115 points. Its average weight was only 6.1 percent, but its own TFP decline was much larger at 1.93 percent per year.

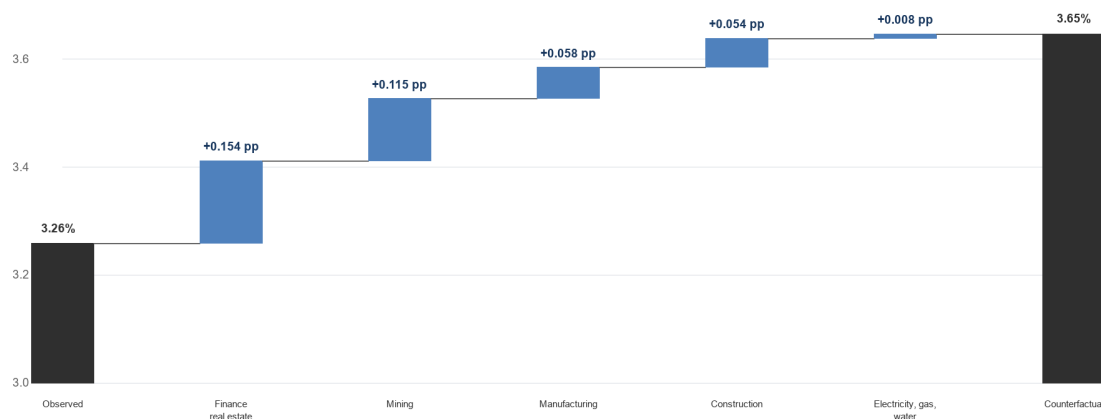
Manufacturing and construction subtracted 0.058 and 0.054 percentage points per year, respectively. Manufacturing had the largest average weight, 23.4 percent, but its own TFP fell at an annualized rate of 0.26 percent. Construction had an average weight of 8.6 percent and a larger sectoral decline of 0.59 percent. These cases show why rankings based only on sectoral TFP growth do not identify the activities with the largest aggregate effect.

2.4 A zero-TFP counterfactual raises production-account output growth

Figure 5 shows the accounting counterfactual. Production-account output grew by 3.26 percent per year between 2005 and 2024. Setting annual TFP growth to zero in the five activities with negative long-run TFP growth raises the average to 3.65 percent. The difference is 0.389 percentage points per year.

Observed and counterfactual production-account output growth

2005–2024; counterfactual sets annual TFP growth to zero in five activities



Note: the vertical scale starts at 3.0%. All other input contributions and annual weights remain fixed.
Source: authors' calculations based on DANE, 2025 TFP annex.

Figure 5: Observed and counterfactual production-account output growth, 2005–2024

Finance and real estate and mining account for 0.269 of the 0.389 additional points. Setting finance and real estate TFP growth to zero adds 0.154 points per year, and doing the same for mining adds 0.115. Manufacturing adds 0.058 points, construction adds 0.054,

and electricity, gas, and water adds 0.008. These changes equal the negative of their average TFP contributions because the exercise holds all other components fixed.

The counterfactual raises output growth in 17 of the 20 years. It lowers growth in 2005, 2006, and 2009 because the five selected activities made a positive joint TFP contribution in those years. Compounding the annual differences implies a production-account output level in 2024 that is 8.1 percent above the observed level.

This result should not be interpreted as a policy estimate. It shows the amount of observed growth associated with the negative TFP contributions under the KLEMS identity. It does not show that zero TFP growth was feasible, which intervention could have achieved it, or how factor use and relative prices would have responded.

3 Discussion

Aggregate stagnation is real, but not a common sectoral outcome. Colombia’s production-account TFP was nearly flat over 2005–2024 because positive contributions from social services, trade, agriculture, and transport were almost fully offset by finance and real estate, mining, construction, manufacturing, and electricity. The relevant empirical question is therefore not only why aggregate TFP stagnated. It is why productivity gains persisted in some activities while negative contributions remained large in others.

The results do not identify the mechanisms behind each activity path. Mining illustrates the limitation. Its output grew by 1.80 percent per year, but the measured input contributions exceeded that growth and produced annualized TFP growth of -1.93 percent. This pattern may reflect resource depletion, the composition of extraction, investment cycles, capacity utilization, regulation, or measurement. The data used here cannot distinguish among these explanations.

Finance and real estate present a different problem. The combined aggregate covers financial intermediation, real estate, and business and rental activities. Its negative contribution may therefore combine distinct subactivities. Output and price measurement are also difficult in financial and real-estate services. A useful extension would reconstruct the contribution at a finer level before assigning a common mechanism to the aggregate.

Manufacturing and construction show why both the sectoral change and the aggregation weight matter. Manufacturing had a moderate TFP decline but a large average weight. Construction had a sharper decline and a smaller weight. Their average aggregate contributions were similar in magnitude. A policy diagnosis based only on the largest sectoral decline would miss manufacturing’s aggregate relevance.

The counterfactual provides a scale for these negative contributions, not a policy target. Raising sectoral TFP requires changes in technology, organization, regulation, factor allocation, capacity use, or measurement, depending on the activity. A credible policy exercise must replace the zero-TFP benchmark with attainable sector-specific references and allow labor, capital, intermediate inputs, prices, and activity weights to respond.

The next empirical step should therefore be sector-specific. For manufacturing, the evidence should test the roles of technology adoption, management practices, exporting, and firm reallocation. For construction, it should distinguish project-cycle and capacity-utilization effects from persistent weaknesses in project management and production meth-

ods. For mining, it should separate resource and composition effects from investment and regulatory conditions. For finance and real estate, the first requirement is a more detailed statistical decomposition.

The analysis also has two general limitations. First, the period begins in 2005 because the sectoral production-account history in the annex is reported as annual changes rather than a longer published index. Second, the public TFP annex does not expose the annual aggregation weights. The recovered weights reconcile the published components exactly and coincide with the Törnqvist averages constructed from DANE’s nominal gross-output series, up to rounding, for 2006–2024 (DANE, 2020, 2026b). Direct validation is unavailable for 2005 because the public base-2015 gross-output series does not include 2004. The accompanying calculation workbook reports the annual comparisons and the treatment of 2005. Replication therefore validates the aggregation weights but not every stage of DANE’s source-data construction.

4 Conclusion

Aggregate TFP stagnation in Colombia over the 20 annual growth rates from 2005 through 2024 was not a common outcome across economic activities. The chained index for the total economy fell from 100 in 2004 to 98.5 in 2024, a cumulative decline of 1.5 percent and an annualized log growth rate of -0.073 percent. Trade TFP increased by 23.2 percent, while mining TFP fell by 32.1 percent. The aggregate result therefore conceals activities moving in opposite directions.

The near-zero total was the balance of contributions with opposite signs. Trade, social services, agriculture, and transport contributed 0.315 percentage points per year. Finance and real estate, mining, construction, manufacturing, and electricity subtracted 0.389 points. Finance and real estate and mining made the largest negative contributions, while trade made the largest positive contribution.

The year-by-year aggregation is central to this conclusion. Cumulative sectoral TFP growth does not add across activities, and average weights applied to long-run sectoral growth do not reproduce the exact total when weights change over time. Recovering the annual weights and averaging annual contributions preserves the published aggregate identity.

The accounting counterfactual shows the scale of the negative contributions. If annual TFP growth in the five activities with negative long-run performance had been zero, production-account output growth would have been 3.65 percent per year rather than 3.26 percent. The 0.39-point difference is economically meaningful, but it is not a causal estimate or a forecast of GDP. The result identifies where a more detailed diagnosis may have the highest aggregate payoff. It does not identify the policies that would produce those gains.

A Average Growth-Accounting Decomposition

Table 2: Average annual decomposition of output growth by activity, 2005–2024

Activity	Output	Labor	Capital	Energy	Materials	Services	TFP
Total economy	3.26	0.92	0.92	0.16	0.60	0.73	-0.07
Agriculture	2.61	-0.07	0.62	0.06	0.89	0.12	0.99
Mining	1.80	0.65	2.95	0.15	0.02	-0.03	-1.93
Manufacturing	2.03	0.21	0.73	0.19	1.07	0.08	-0.26
Electricity, gas, and water	2.66	0.31	1.22	1.11	0.10	0.21	-0.28
Construction	2.27	0.72	0.80	0.01	1.34	-0.02	-0.59
Trade, hotels, and restaurants	3.79	0.86	0.16	0.09	0.62	1.01	1.04
Transport and communications	4.28	1.32	0.73	0.53	0.18	1.32	0.21
Finance and real estate	3.95	1.72	2.09	0.01	0.05	1.14	-1.05
Social services	4.69	1.78	0.07	0.04	0.37	1.76	0.66

Notes: Values are average annual percentage-point contributions, except output and TFP, which are annualized log growth rates. The 20 observations from 2005 to 2024 connect the levels in 2004 and 2024. The total-economy row is the DANE published aggregate, not a simple sum or average of the activity rows. Components may not add exactly because of rounding.

Source: Authors' calculations based on DANE, 2025 TFP annex.

B Contributions by Subperiod

Table 3: Activity contributions to aggregate TFP by subperiod

Activity	2005–10	2011–15	2016–19	2020–24	2005–24
Agriculture	-0.069	0.122	0.094	0.110	0.056
Mining	0.009	-0.193	-0.155	-0.153	-0.115
Manufacturing	-0.022	-0.157	-0.021	-0.031	-0.058
Electricity, gas, and water	-0.006	-0.035	-0.018	0.026	-0.008
Construction	0.059	-0.022	-0.078	-0.202	-0.054
Trade, hotels, and restaurants	0.085	0.216	0.077	0.140	0.130
Transport and communications	-0.036	0.076	0.038	0.005	0.017
Finance and real estate	-0.224	-0.038	-0.169	-0.173	-0.154
Social services	-0.030	0.054	0.192	0.276	0.112
Total economy	-0.235	0.022	-0.040	-0.001	-0.073

Notes: Average annual percentage-point contributions. Each activity entry is the subperiod mean of annual TFP multiplied by the annual aggregation weight. Subperiods have different lengths.

Source: Authors' calculations based on DANE, 2025 TFP annex.

C Activity Coverage

Table 4: Correspondence between KLEMS aggregates and CIU activities

Activity aggregate	Included CIU activities
Agriculture	Section A: agriculture, livestock, hunting, and forestry. Section B: fishing.
Mining	Section C: mining and quarrying.
Manufacturing	Section D: manufacturing, including food and beverages; tobacco; textiles and apparel; leather and footwear; wood; paper and printing; petroleum refining; chemicals; rubber and plastics; non-metallic mineral products; metals; machinery and equipment; electrical, electronic, and transport equipment; and other manufacturing.
Electricity, gas, and water	Section E: electricity, gas, and water supply.
Construction	Section F: construction.
Trade, hotels, and restaurants	Section G: wholesale and retail trade; repair of motor vehicles, motorcycles, personal effects, and household goods. Section H: hotels and restaurants.
Transport and communications	Section I: transport, storage, and communications.

Activity aggregate	Included CIIU activities
Finance and real estate	Section J: financial intermediation. Section K: real estate, renting, and business activities.
Social services	Section L: public administration and defense; compulsory social security. Section M: education. Section N: health and social work. Section O: other community, social, and personal service activities. Section P: private households with employed persons and undifferentiated production activities of private households. Section Q: extraterritorial organizations and bodies.

Notes: The table documents the correspondence between the nine aggregates in the DANE annex and the CIIU Rev. 3/3.1 sections adapted for Colombia. Labels may vary across revisions without changing the KLEMS aggregate used by DANE.

Source: DANE, CIIU Rev. 3 and Rev. 3.1 A.C.; 2025 TFP annex.

Data and Replication Files

The DANE source annexes, processed data, replication programs, figures, and table sources are available in the project repository:

https://github.com/oliverpardo1979/informe_ptf.

The spreadsheet calculations behind the six figures and the paper’s principal results are contained in the six sheets whose names begin with CJC in the [primary TFP calculation workbook](#). Those sheets report the aggregate value-added decomposition, chained activity indices, annual gross-output weights, sectoral contributions, counterfactuals, and the source range for each figure. The sheet **CJC Guía** provides a map of the calculations and distinguishes source data from formulas.

The same six calculation sheets are also included in the [companion labor-productivity workbook](#). This second file is a convenience copy, not an independent source of calculations or results: the two workbooks retain different original DANE tables but share the same CJC calculation package. The original DANE annexes are redistributed in the repository for reproducibility; users should consult DANE for the official releases and any subsequent revisions.

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